Word2Vec *using* IITJ Institutional Corpus

CSL 7640: Natural Language Understanding

Assignment 2 — Problem 1

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Abstract

Word embeddings are dense vector representations that capture semantic and syntactic relationships between words in a corpus. In this project, word embeddings are learned from a collection of institutional documents related to IIT Jodhpur. Two variants of the Word2Vec algorithm, Continuous Bag of Words (CBOW) and Skip-gram with Negative Sampling, are implemented and evaluated. The models are trained under multiple hyperparameter configurations and evaluated using nearest-neighbor similarity queries, analogy reasoning tasks, and embedding visualizations using PCA and t-SNE. Experimental observations indicate that Skip-gram embeddings capture semantic relationships more effectively for the given dataset. Limitations of the corpus size and potential improvements for future work are also discussed.

1 Introduction

Word embeddings provide a powerful mechanism for representing words in a continuous vector space where semantic relationships between words can be captured geometrically. Traditional one-hot word representations treat words as independent symbols and fail to encode relationships between them. Word2Vec models address this limitation by learning distributed representations based on contextual co-occurrence patterns in large text corpora.

In this assignment, word embeddings are learned from textual data derived from IIT Jodhpur institutional documents such as reports, academic descriptions, and administrative material. The objective is to investigate whether meaningful semantic relationships can be captured from a domain-specific corpus.

Two Word2Vec architectures are implemented and compared:

- Continuous Bag of Words (CBOW)

- Skip-gram with Negative Sampling

The learned embeddings are evaluated using similarity queries, analogy reasoning, and visualization of the embedding space.

2 Dataset

2.1 Data Collection

s The dataset consists of institutional documents related to IIT Jodhpur collected from publicly available sources. These documents include academic program descriptions, institutional reports, research summaries, and administrative materials.

The documents were converted into plain text and combined into a single corpus file used for training the embedding models.

2.2 Characteristics

The corpus reflects academic and institutional language. Frequently occurring terms include words related to education, research, academic programs, and institutional infrastructure.

2.3 Preprocessing

Before training the Word2Vec models, the text undergoes a preprocessing pipeline to remove noise and standardize the input. The following preprocessing steps are applied:

- Conversion of text to lowercase
- Removal of URLs, HTML tags, and formatting artifacts
- Tokenization using the spaCy library
- Removal of punctuation and non-alphabetic tokens
- Stopword filtering

Additionally, words occurring fewer than five times in the corpus are removed from the vocabulary to reduce noise and improve training efficiency.

2.4 Corpus Visualization

To obtain an intuitive understanding of the most frequent words in the dataset, a word cloud visualization was generated.

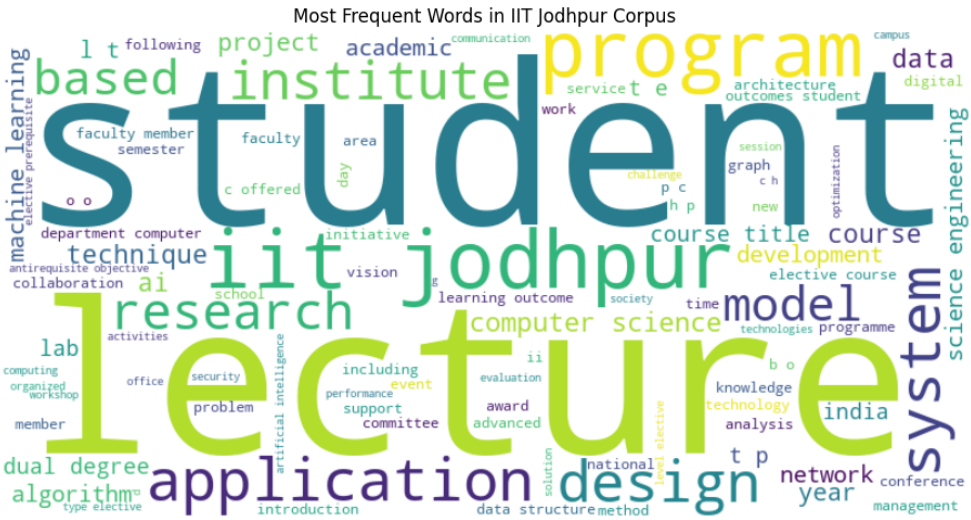


Figure 1: Word cloud showing frequent terms in the IIT Jodhpur corpus

The visualization shows that terms such as *student*, *institute*, *research*, *program*, and *system* dominate the corpus, reflecting the academic nature of the dataset.

3 Word2Vec Model Architecture

Word2Vec learns word embeddings by predicting contextual relationships between words in a corpus. Two architectures are implemented in this project.

3.1 Continuous Bag of Words (CBOW)

The CBOW model predicts a target word given its surrounding context words.

Given a context window:

$$w_{t-c}, ..., w_{t-1}, w_{t+1}, ..., w_{t+c}$$

the model predicts the probability: $P(w_t|context)$

The context word embeddings are averaged and passed through a linear layer followed by a softmax function.

3.2 Skip-gram Model

The Skip-gram model performs the reverse operation by predicting surrounding context words given the target word.

The training objective is:

$$\sum_{t=1}^{T} \sum_{|j| \leq c, j \neq 0} \log P(w_{t+j} | w_t)$$

Skip-gram is known to produce better representations for rare words and smaller datasets.

3.3 Negative Sampling

Computing the full softmax over large vocabularies is computationally expensive. Negative sampling is used to approximate the objective function.

For a word-context pair (w, c), the objective becomes:

$$\log \sigma(v_c^T v_w) + \sum_{i=1}^{k} \log \sigma(-v_{n_i}^T v_w)$$

where n_i are negatively sampled words.

3.4 Evaluation Pipeline

After training, the learned embeddings were evaluated using:

- Cosine similarity based nearest neighbor queries
- Vector arithmetic based analogy reasoning
- Two-dimensional visualization using PCA and t-SNE

These evaluation methods provide both qualitative and structural insights into the learned semantic representations.

3.5 Software and Tools

The implementation relied on several Python libraries:

- **PyTorch** for neural network training
- **spaCy** for text tokenization
- **NumPy** for numerical operations

- **Matplotlib** for visualization
- **scikit-learn** for PCA and t-SNE dimensionality reduction

4 Experimental Setup

Multiple hyperparameter configurations were tested to analyze their effect on embedding quality.

Table 1: Used Hyperparameter Values

Embedding Dimension	Window Size	Negative Samples
50	2	5
100	5	10
300	8	15

Here each hyperparameter was used for all possible combinations. Models were trained using stochastic gradient descent for multiple epochs.

5 Semantic Evaluation

To evaluate the quality of the learned embeddings, cosine similarity was used to retrieve the nearest neighbors for selected query words.

Example queries included:

- research
- student
- phd

For Skip-gram models, nearest neighbors for *research* included terms such as *interdisciplinary*, *areas*, and *innovation*, demonstrating meaningful semantic associations.

Analogy reasoning tasks were also performed. For example:

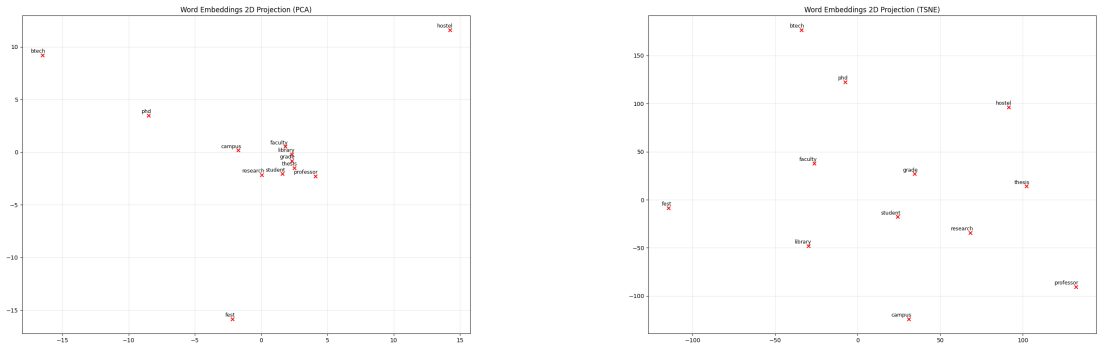
- UG : BTech :: PG : MTech
- Culture : Ignus :: Sports : Varchas

These results indicate that the embedding space captures certain semantic relationships between institutional data.

6 Embedding Visualization

To analyze the structure of the embedding space, dimensionality reduction techniques were applied to project the vectors into two dimensions.

6.1 CBOW Embeddings

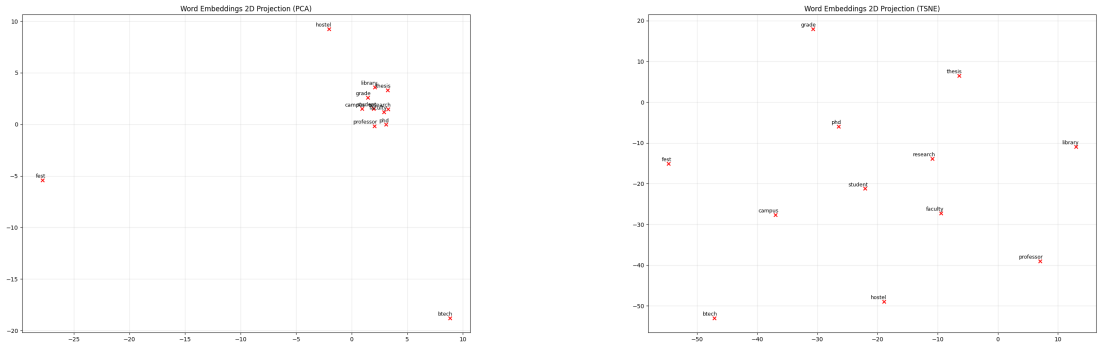


(a) PCA projections (b) t-SNE projections

Figure 2: CBOW word embeddings

The CBOW embeddings show moderate clustering of related academic terms.

6.2 Skip-gram Embeddings



(a) PCA projections (b) t-SNE projections

Figure 3: Skip-gram word embeddings

Compared to CBOW, the Skip-gram model produces clearer semantic groupings between related words such as research-related terms and campus-related vocabulary.

7 Discussion

The experimental results suggest that Skip-gram embeddings capture semantic relationships more effectively than CBOW for the IIT Jodhpur corpus. This observation is consistent with prior research, which shows that Skip-gram performs better on smaller datasets and for capturing rare word relationships.

However, the relatively small corpus size limits the richness of the learned embeddings. Some analogy predictions remain noisy, and certain words do not appear in the vocabulary due to frequency filtering.

8 Results Summary

A qualitative comparison of CBOW and Skip-gram models indicates that Skip-gram consistently produces more semantically meaningful embeddings for the IIT Jodhpur corpus.

Model	Strengths	Weaknesses
CBOW	Faster training	Less accurate semantic relationships
Skip-gram	Better rare word representations	Slightly slower training

Table 2: Comparison of CBOW and Skip-gram models

Skip-gram models particularly perform better for analogy tasks and nearest-neighbor semantic similarity queries.

9 Conclusion

This project explored the training of Word2Vec embeddings using a domain-specific corpus derived from IIT Jodhpur institutional documents. Both CBOW and Skip-gram models were implemented and evaluated across multiple hyperparameter configurations.

Results from similarity queries, analogy tasks, and embedding visualizations indicate that Skip-gram models produce more meaningful representations for the given dataset. Despite the limited corpus size, the learned embeddings capture several useful semantic relationships.

Future work could involve expanding the corpus with research publications, incorporating subword models such as FastText, or evaluating the embeddings in downstream NLP tasks.